Release Plan

StageOne: Indie-Musician Discovery Platform

WEB 268 Capstone Project

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1. Executive Summary

This document describes the end-to-end release plan for StageOne, a full-stack MERN application that helps indie musicians publish profiles, share tracks and gigs, and connect with fans. The release plan defines how the development team will move StageOne from an empty repository to a publicly reachable, client-approved production deployment.

The plan is structured around five incremental, testable releases, each mapped to a specific set of academic milestones. Every release ships through a three-environment promotion pipeline (Development > Staging > Production) with automated CI/CD gates, accessibility checks, and a documented rollback path.

Total estimated implementation effort is 372 developer-hours. At a commercial blended rate of $85/hr, this represents a commercial-equivalent implementation cost of $31,620. First-year hosting, domain, email, and third-party service costs total $480.84 at commercial rates. However, because StageOne is an academic capstone executed by a single student developer on free-tier infrastructure, the actual out-of-pocket cost to the client is $0.00!

2. Project Overview

2.1 Scope in one paragraph

StageOne is a public-facing web application and administrative back-office for independent musicians. The public experience allows fans to browse artist profiles, stream track previews, follow artists, and view upcoming gigs. The authenticated artist experience allows musicians to manage their profile, upload tracks and gig announcements, and view follower growth. An AI concierge (powered by OpenAI with a local Ollama fallback) answers natural-language discovery questions such as 'what's a good show this Friday?' The stack is MongoDB Atlas, Express.js on Render, React on Vercel, and Cloudinary for media.

2.2 Release-plan definitions

Release: A numbered version of StageOne that crosses a quality gate and is deployed to at least the Staging environment.

Deployment: The act of promoting a release artifact from one environment to the next.

Cutover: The one-time event when the production URL begins serving live traffic.

Hypercare: The three days immediately following cutover during which the developer is on-call for P0/P1 issues.

Handover: The formal transition to the client, at which point ownership of hosting credentials, admin accounts, and documentation passes from the developer to the client.

2.3 Client success criteria

- The production site is reachable via HTTPS at a client-owned custom domain.
- A fan can sign up, follow an artist, and receive an email confirmation.
- An artist can log in, edit their profile, upload a track with cover art, and publish a gig.
- Lighthouse scores ≥ 90 for Performance, Accessibility, Best Practices, and SEO on mobile.
- All client-owned credentials are transferred to the client in a documented handover packet.
- The client signs the Section 13 sign-off block of this document acknowledging completion.

3. Release Strategy

3.1 Release cadence

StageOne follows a phased, incremental release strategy. Rather than a single big-bang launch, the team ships five named releases over the 15-week window. Each release is end-to-end functional — that is, it can be demoed to the client as a working, if minimal, product — and each release closes a defined quality gate before work on the next begins.

Release	Target date	Headline capability	Environment reached
v0.1 (Skeleton)	February 4, 2026	Hello-world React app calling Express API; MongoDB Atlas connected; CI pipeline green.	Dev only
v0.2 (Auth & profiles)	February 25, 2026	JWT login/registration; artist and fan profiles; role-based access.	Dev + Staging
v0.3 (Content & discovery)	March 22, 2026	Track and gig CRUD with Cloudinary uploads; browse pages; follow/unfollow.	Dev + Staging
v1.0-rc (Release candidate)	April 20, 2026	AI assistant, accessibility and performance passes, seed data, client UAT build.	Dev + Staging
v1.0 (Production launch)	April 30, 2026	Production cutover at 10:00 a.m. ET; hypercare begins.	Production

3.2 Branching and source-control model

main: Always deployable. Protected; direct pushes disabled; merges only through approved pull requests that pass CI.

develop: Integration branch. All feature branches merge here first; automatically deployed to the Dev environment on green CI.

feature/*: Short-lived feature branches (ideally ≤ 3 days old) cut from develop and merged back.

release/x.y: Cut from develop when a release is locked. Only bugfixes and docs land here; merged to both main and develop at release time.

hotfix/*: Cut from main for any P0/P1 issue discovered in production during hypercare; merged back to main and develop.

3.3 Hosting approach

StageOne uses free-tier, managed third-party hosts rather than a client-owned server. This choice is explicit: it removes the infrastructure burden from the client, reduces first-year cost to near zero, and lets the developer focus on application code rather than server administration. The four service providers are MongoDB Atlas (database), Render (Node API), Vercel (React static build and CDN), and Cloudinary (image, audio, and video delivery). Render and Vercel both provide zero-downtime deploys on push-to-main, which eliminates the need for a separate blue/green infrastructure layer for the capstone's traffic profile.

4. Environments

StageOne uses three isolated environments. Each environment has its own database cluster, its own API instance, its own frontend build, its own Cloudinary folder, and its own environment variables. No secrets are ever shared across environments.

Development (dev)

Feature authoring and local smoke tests. Runs on the developer's laptop and in short-lived PR preview deployments. A lightweight seeded fixture (5 artists, 15 tracks, 8 gigs). Wiped freely.

Access: Developer only. PR previews gated by an auth basic-auth wall.

Cost: $0 (local) / free Render and Vercel PR previews.

Staging

Client demos, accessibility audits, and pre-release regression. Must mirror production configuration except for the domain. The production seed dataset plus scripted test accounts. Refreshed weekly on Sunday night.

Access: Developer + client (instructor). Behind a login wall with a test-user account.

Cost: $0 (Render free web service + Vercel hobby tier).

Production

Live, client-approved public instance. Real artist submissions once launched; initial seed of 10 artists provided by the developer.

Access: Public read; role-gated write.

Cost: See Sections 10 and 11.

4.1 Environment-variable matrix

The table below documents the environment-variable contract across the three environments. Every variable that appears in any environment must appear in all three — missing variables cause a deploy-time check to fail in CI.

Variable	Dev	Staging	Production
NODE_ENV	development	staging	production
MONGODB_URI	dev cluster URI	staging cluster URI	prod cluster URI
JWT_SECRET	dev-only	staging-only	prod-only (32-byte)
CLOUDINARY_URL	dev folder	staging folder	prod folder
OPENAI_API_KEY	personal	shared-staging	shared-prod (rate-limited)
OLLAMA_HOST	localhost:11434	staging sidecar	prod sidecar
RESEND_API_KEY	test mode	live (staging domain)	live (prod domain)
SENTRY_DSN	(blank)	staging DSN	prod DSN
CORS_ORIGIN	http://localhost:5173	https://staging.stageone.app	https://www.stageone.app

5. Work-Breakdown Structure (Implementation Hours)

The table below breaks the 372-hour implementation effort into twelve work packages. Hours are estimates for a single developer working part-time (~24 hrs/week) over the 15-week window. A 15% contingency buffer is embedded within individual packages rather than called out as a separate line, which is why some estimates appear slightly rounded-up.

#	Work package	Key deliverables	Hours
1	Project setup	Repo, monorepo structure, linting, Prettier, Husky, commitlint, CI skeleton.	12
2	Data modeling	ERD, Mongoose schemas for User, Artist, Track, Gig, Follow, Message; seed script.	24
3	Auth & RBAC	Registration, login, JWT + refresh, bcrypt hashing, role middleware (Fan/Artist/Admin).	36
4	API endpoints	RESTful routes for all six resources, validation (Zod), pagination, error handling.	48
5	Frontend shell	Vite + React + TypeScript, router, layout, auth context, design-token CSS variables.	24
6	Public pages	Home, Discover, Artist page, Track page, Gig detail, Search, 404.	52
7	Authenticated pages	Artist dashboard, profile edit, track upload, gig creation, messages inbox.	48

#	Work package	Key deliverables	Hours
8	Media pipeline	Cloudinary signed uploads, audio player, image cropping/transform, CDN headers.	24
9	AI assistant	OpenAI function-calling, local Ollama/Llama-3 fallback, prompt library, rate limiting.	28
10	A11y + performance	WCAG 2.1 AA audit, color-contrast pass, Lighthouse fixes, image lazy-loading, bundle split.	20
11	Launch ops	Domain, DNS, TLS, monitoring, backups, runbooks, pre-launch checklist execution.	28
12	Documentation & handover	README, API reference, admin guide, 30-minute handover video, this release plan.	28
	Total implementation effort		**372**

6. Phased Schedule (Dates & Durations)

The 15-week window is partitioned into six phases. Each row below gives the phase name, start and end dates, the calendar duration, the target number of developer hours within the phase, and the release artifact that closes it.

#	Phase	Dates	Duration	Hours
1	Discovery & design	Mon Jan 19 – Sun Feb 1, 2026	2 weeks	48
2	Foundation build	Mon Feb 2 – Sun Feb 22, 2026	3 weeks	72
3	Core features	Mon Feb 23– Sun Mar 22, 2026	4 weeks	96
4	Polish & AI	Mon Mar 23 – Sun Apr 19, 2026	4 weeks	88
5	UAT & launch prep	Mon Apr 20 – Thu Apr 30, 2026	11 days	44
6	Cutover & hypercare	Fri May 1 – Mon May 4, 2026	4 days	24
	Total	Jan 20 – May 4, 2026	15 weeks	**372**

6.1 Weekly cadence within each phase

Each week follows a fixed rhythm: Monday planning (30 min), Tuesday–Thursday deep work (3 × 6 hr sessions), Friday demo-and-review with the client (45 min), and Sunday retrospective + triage (60 min). This totals roughly 24 developer-hours per week, which matches the 372-hour bottom-line across 15 weeks with a ~5% slack factor built in.

7. Pre-Launch Checklist

The following forty-checkpoint list is adapted from items 31 through 55 of Andy Crestodina's Website Launch Checklist (Orbit Media Studios). Each item is mapped to a due date and verification method. The checklist is executed in Phase 5 (Apr 20 – Apr 30, 2026) and must be 100% green before the go/no-go meeting on Apr 30.

#	Checkpoint	Due	Verification
31	Favicon uploaded and rendered in tabs/ bookmarks.	Apr 24	Visual check across Chrome, Safari, Firefox
32	Open Graph + Twitter Card meta tags set for every page type.	Apr 24	opengraph.xyz + Twitter card validator
33	Custom 404 page with navigation back to Home.	Apr 24	Manual test /does-not-exist
34	Custom 500 page with generic error + support link.	Apr 24	Force error in staging; visual check
35	SSL/TLS certificate valid (A or A+ on SSL Labs).	Apr 24	ssllabs.com scan
36	HSTS header set with preload.	Apr 24	securityheaders.com scan
37	Mixed-content warnings = 0.	Apr 24	Chrome DevTools Security panel
38	robots.txt present and correct for prod vs staging.	Apr 24	Direct fetch
39	sitemap.xml auto-generated and linked from robots.txt.	Apr 24	xml-sitemaps.com validator
40	Google Search Console property verified; sitemap submitted.	Apr 24	Search Console dashboard
41	Google Analytics 4 property installed; events firing.	Apr 27	GA4 DebugView
42	Cookie consent banner.	Apr 27	Visual + localStorage check
43	Privacy policy page linked in footer.	Apr 27	Visual check
44	Terms of service page linked in footer.	Apr 27	Visual check
45	Contact page with working form; submissions arriving.	Apr 27	Live test submit
46	Email deliverability verified (SPF + DKIM + DMARC).	Apr 27	mail-tester.com ≥ 9/10
47	Transactional email templates proofread.	Apr 27	Client review
48	404 rate < 1% on staging soak test.	Apr 28	Sentry + server logs
49	Uptime monitor configured (5-min interval).	Apr 28	Dashboard screenshot
50	Error monitoring configured, source maps uploaded.	Apr 28	Force error in staging
51	Backups configured; nightly Atlas snapshots.	Apr 29	Atlas backup dashboard
52	Restore-from-backup drill executed at least once.	Apr 29	Runbook with timestamps
53	Performance: Lighthouse Mobile ≥ 90.	Apr 29	Lighthouse HTML report

#	Checkpoint	Due	Verification
54	Accessibility: axe-core scan = 0 critical/serious issues.	Apr 29	axe DevTools export
55	Cross-browser check.	Apr 30	BrowserStack screenshots

All forty items above are tracked as individual GitHub Issues in the StageOne repo under the 'launch-checklist' project board. The board is reviewed at the daily stand-up during Phase 5 and must reach 100% closed status by Jul 30 5:00 p.m. ET.

8. Deployment & Cutover Plan

8.1 Standard deployment (any release ≤ v1.0-rc)

Day-to-day deployments are fully automated. A merge to main triggers the following sequence:

1. CI runs unit tests, lint, type-check, and axe-core accessibility scan. Any failure aborts the deploy.
2. CI builds the React bundle and the Express server.
3. Vercel publishes the static bundle to its CDN.
4. Render performs a rolling deploy of the Node service — new instance is created, health-checked, then traffic is flipped.
5. A post-deploy smoke script hits critical endpoints. Any 5xx response triggers Sentry alert + Slack ping.

8.2 Production cutover (May 1, 2026, 10:00 a.m. ET)

The one-time promotion from the staging domain to the production domain follows a stricter script. All times are Eastern Time.

Time	Step	Exit criterion
T-24 hrs (Apr 30 10:00)	Final regression pass on staging.stageone.app	Zero P0/P1 bugs open
T-18 hrs (Apr 30 16:00)	Go/no-go meeting with client	Written go decision
T-12 hrs (Apr 30 22:00)	DNS TTL pre-lowered to 300s	dig confirms TTL
T-60 min (09:00)	Atlas snapshot taken + labeled 'pre-launch-v1.0'	Snapshot visible in Atlas UI
T-30 min (09:30)	Freeze merges to main branch	GitHub branch rule toggled
T-15 min (09:45)	Announce freeze in #stageone-launch	Message posted
T-0 (10:00)	Tag v1.0; promote to prod; switch DNS A record for www to Vercel	www.stageone.app returns 200 OK
T+5 min (10:05)	Smoke script run against production	All 5 checks pass

Time	Step	Exit criterion
T+10 min (10:10)	Lighthouse run on production home page	All ≥ 90
T+15 min (10:15)	Client smoke test — client logs in, edits a profile	Client confirms success
T+30 min (10:30)	Public announcement post	Post live
T+60 min (11:00)	Freeze lifted; hypercare monitoring begins	Branch rule restored

8.3 Rollback plan

If smoke-test step T+5 fails or if the client's T+15 check fails, the developer executes the rollback procedure below. The target is to restore www.stageone.app to a last-known-good state within 15 minutes.

1. Flip the Vercel production alias back to the previous deployment. ETA: 30 seconds.
2. Roll back the Render service to the previous immutable image tag. ETA: 3 minutes.
3. If a schema change was shipped: run the down-migration script from the release's migrations/ folder. ETA: 5 minutes.
4. If data was corrupted: restore the pre-launch-v1.0 Atlas snapshot into a new cluster and swap the connection string. ETA: 10 minutes.
5. Announce rollback in #stageone-launch; file a P0 issue with the reason.
6. Schedule a post-mortem within 48 hours.

9. Post-Launch Support Plan

Post-launch support spans the 72-hour hypercare window immediately after cutover and a 90-day warranty period.

9.1 Hypercare

On-call hours: 24/7. Developer reachable via phone and email.

P0 SLA: Acknowledge within 1 hour; mitigation within 4 hours.

P1 SLA: Acknowledge within 4 hours; mitigation within 1 business day.

Daily stand-up: 15-minute sync with client every day at 9:00 a.m. ET.

Exit criterion: Zero open P0/P1 issues for 24 contiguous hours.

9.2 Warranty period

Covered: Bug fixes on functionality that was working at launch; documentation clarifications; minor accessibility or SEO adjustments.

Not covered: New features; scope changes; content entry; third-party outages.

Response time: Best-effort within 3 business days, as developer availability is constrained by the fall academic term.

Cap: 15 developer hours total across the 90-day window. Additional effort would be scoped as a paid change request at the rate stated in **Section 10.2**.

9.3 Handover artifacts

- Credentials packet containing: domain registrar, Vercel, Render, Atlas, Cloudinary, Resend, Sentry, UptimeRobot, GA4, Search Console.
- Admin user guide covering login, content moderation, and the Atlas backup dashboard.
- Developer README and API reference in the GitHub repository.
- Pre-recorded 30-minute walk-through video on the client's preferred video host.

10. Implementation Cost Breakdown

This section presents two parallel views: (a) the commercial-equivalent cost, which is what a freelance agency or full-time employee at market rate would charge to deliver the same scope, and (b) the actual cost to the client (WEB 268 instructor), which is zero because StageOne is an academic capstone produced by an enrolled student.

10.1 Labor — commercial-equivalent

Blended developer rate of $85.00/hr reflects a mid-level full-stack contractor in the Harrisburg, PA market as of April 2026. Individual rates vary by work package — for example, design work at $70/hr, backend development at $85/hr, and DevOps/launch work at $110/hr — but for simplicity the blended rate is applied uniformly below, and the per-package variances net out within 5% of the blended total.

#	Work package	Hours	Rate	Subtotal
1	Project setup	12	$85.00 / hr	$1,020.00
2	Data modeling	24	$85.00 / hr	$2,040.00
3	Auth & RBAC	36	$85.00 / hr	$3,060.00
4	API endpoints	48	$85.00 / hr	$4,080.00
5	Frontend shell	24	$85.00 / hr	$2,040.00
6	Public pages	52	$85.00 / hr	$4,420.00
7	Authenticated pages	48	$85.00 / hr	$4,080.00
8	Media pipeline	24	$85.00 / hr	$2,040.00
9	AI assistant	28	$85.00 / hr	$2,380.00
10	A11y + performance	20	$85.00 / hr	$1,700.00
11	Launch ops	28	$85.00 / hr	$2,380.00

#	Work package	Hours	Rate	Subtotal
12	Documentation & handover	28	$85.00 / hr	$2,380.00
	Labor subtotal	**372**		**$31,620.00**

10.2 One-time non-labor implementation costs

Line item	Notes	Cost
Domain registration	Porkbun, 1-year initial term	$13.99
Figma Professional subscription (3mo)	Required for wireframes and color comps	$45.00
Stock imagery + audio for seed data	Unsplash+ / Pixabay Pro (1mo)	$29.00
SSL Labs paid scan credits	Optional; replaced by free scans in practice	$0.00
Initial BrowserStack (1mo) for cross-browser QA	Used for checklist item #55	$39.00
One-time misc. (buffer, 5%)	Typos in docs, small fees	$25.00
Non-labor subtotal		**$151.99**

10.3 Implementation grand totals

Scenario	Labor	Non-labor	Total
Commercial-equivalent (agency / contractor)	$31,620.00	$151.99	$31,771.99
Actual (academic capstone)	$0.00	$0.00	$0.00

The 'actual' row is zero because the developer is an enrolled student working within the WEB 268 capstone course — the labor is already paid for by the student's tuition and HACC's course infrastructure. The $151.99 of non-labor implementation costs is absorbed by the developer as part of their normal coursework tools-and-materials budget.

11. First-Year Operating Cost Breakdown

This section projects ongoing costs from production launch on Apr 30, 2026 through Apr 30, 2027. Again two views are provided: the commercial-equivalent cost (if the project were hosted on paid tiers suitable for a small-business client) and the actual cost the client incurs during year one (which leverages free tiers where possible).

11.1 Commercial-equivalent monthly recurring cost

Service	Tier	Monthly	Annual
MongoDB Atlas	M10 dedicated (2 GB, 10 GB storage)	$57.00	$684.00
Render Web Service	Standard (1 CPU, 2 GB RAM)	$25.00	$300.00
Vercel	Pro seat	$20.00	$240.00
Cloudinary	Plus plan (30 credits)	$89.00	$1,068.00
Resend (transactional email)	Pro (50K emails / mo)	$20.00	$240.00
Sentry (error monitoring)	Team plan	$26.00	$312.00
UptimeRobot	Pro (1-min interval, SMS alerts)	$7.00	$84.00
OpenAI API	Estimated usage @ ~2M tokens/mo	$12.00	$144.00
Recurring subtotal		**$256.00**	**$3,072.00**

11.2 Commercial-equivalent annual-only items

Line item	Notes	Annual
Domain renewal	Year 2 registration	$13.99
SSL/TLS	Let's Encrypt via Vercel/Render	$0.00
Professional email	hello@stageone.app	$72.00
Business-class DNS (Cloudflare Pro)	Optional; WAF + analytics	$240.00
Security / accessibility audit (annual)	3rd-party penetration + a11y retest	$850.00
Annual subtotal		**$1,175.99**

11.3 Commercial-equivalent first-year total

Component	Amount
Recurring (12 × $256.00)	$3,072.00
Annual-only items	$1,175.99
Commercial-equivalent Year-1 total	**$4,247.99**

11.4 Actual Year-One cost (as deployed)

StageOne's academic deployment uses the free or entry tier of every provider listed above. Since a single-student capstone generates negligible traffic and data volume, these tiers are sufficient. The only out-of-pocket costs are domain registration and a small OpenAI usage allowance.

Service	Tier used	Monthly	Annual
MongoDB Atlas	M0 free cluster	$0.00	$0.00
Render Web Service	Free instance (spin-down after 15 min idle)	$0.00	$0.00
Vercel	Hobby plan	$0.00	$0.00
Cloudinary	Free plan (25 credits)	$0.00	$0.00
Resend	Free tier (100 emails/day)	$0.00	$0.00
Sentry	Developer tier	$0.00	$0.00
UptimeRobot	Free tier (5-min interval)	$0.00	$0.00
OpenAI API	Pay-as-you-go, ~$1/mo at capstone usage	$1.00	$12.00
Domain	Porkbun	—	$13.99
Actual Year-1 total			**$25.99**

Both Year-1 totals are absorbed by the developer out of pocket for the duration of the capstone showcase. If the client later chooses to keep StageOne online past the showcase, the client would either take over the $25.99/yr costs directly (simplest) or migrate to paid tiers per Section 11.1 for a more resilient deployment. The plan formally budgets for the second scenario so there are no surprises for the client.

12. Risk Register

The top eight risks to the release plan are tracked below with likelihood, impact, and the mitigation the team will execute if the risk materializes.

#	Risk	Likelihood	Impact	Mitigation
R1	Scope creep pushes v1.0-rc past Apr 20.	Medium	High	Freeze feature list at end of Phase 2; any new request goes to a v1.1 backlog.
R2	Render cold start produces > 30s first-request latency.	High	Medium	Cron-ping every 10 minutes during waking hours; increase budget tier if it persists.
R3	API outage disables AI assistant on demo day.	Low	Medium	Ollama/Llama-3 local fallback shipped in Phase 4; feature-flag allows disable.
R4	Atlas rate-limit throttles during UAT load.	Low	Medium	Pre-launch load test on Apr 28; upgrade to M2 ($9/mo) if throttled.
R5	DNS propagation delay prevents launch.	Low	High	TTL pre-lowered to 300s 24 hours ahead; domain/CNAME pre-configured.
R6	Developer illness during Phase 5.	Low	High	Two-day schedule buffer built into Phase 5; non-critical checklist items pre-executed.

#	Risk	Likelihood	Impact	Mitigation
R7	Client unavailability during UAT week.	Medium	High	UAT window confirmed with client at project kickoff; back-up reviewer identified.

13. Sign-Off

By signing below, the parties acknowledge that the release plan defined in this document — including the 15-week schedule (Sections 5 and 6), the pre-launch checklist (Section 7), the deployment and rollback procedures (Section 8), the post-launch support commitment (Section 9), and the cost breakdowns (Sections 10 and 11) — is mutually understood and accepted as the basis for the StageOne capstone delivery.

Developer: Eva Quinn Cihak

Signature: ______________________________ Date: ____________

Client: Eric Yoxheimer

Signature: ______________________________ Date: ____________